

Final Exam

Fiscal and Monetary Policy 2026

Sample Exam

Instructions.

- Time: **90 minutes**. Total: **100 points**.
 - The exam has two parts: **Part A** – eight True/False/Uncertain questions (5 points each, 40 points); **Part B** – two short model problems (30 points each, 60 points).
 - For Part A, a verdict (true / false / uncertain) earns no credit on its own: state your verdict and *justify* it with economic reasoning, algebra, or a graph.
 - Show your steps. If you get stuck on algebra, explain the economics and go as far as you can.
 - Open book: notes and printed materials are allowed. No electronic devices.
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Part A. True / False / Uncertain ($8 \times 5 = 40$ points)

State whether each claim is true, false, or uncertain, and justify.

1. Under active fiscal policy, doubling the inherited stock of nominal government debt while holding the path of real primary surpluses fixed doubles the price level.
2. In an active-monetary/passive-fiscal regime a primary-surplus shock does not move inflation, and this insulation survives once agents attach positive probability to a switch to a fiscal-dominant regime.
3. In the neoclassical growth model with inelastic labor supply, a permanent lump-sum-financed increase in government purchases leaves output unchanged and crowds out consumption one-for-one.
4. Nominal rigidities are sufficient to generate a government-spending multiplier greater than one.
5. Under complete markets (Lucas–Stokey), the Ramsey government uses state-contingent debt to smooth the labor-tax distortion; with the preferences used in class the optimal tax rate is the same in every state.
6. When the available non-contingent bonds have a nearly singular return matrix across states, implementing the complete-markets allocation requires only small bond positions.
7. If the interest rate is below the growth rate ($\bar{R} < \bar{\Gamma}$), the government must still run a primary surplus to keep the debt-to-GDP ratio stable.
8. Suppose safe interest rates lie below the economy's growth rate. Then valuing a claim to future GDP at the risk-free rate – i.e. ignoring the GDP risk premium – gives an *infinite* present value, so a finite fiscal capacity relies on the risk premium.

Part B. Two short model problems ($2 \times 30 = 60$ points)

Problem 1. Price-level determination under interest-rate rules (30 points)

Cashless endowment economy: $y_t = y$, $c_t = y$, complete markets, constant real rate, discount factor $\beta \in (0, 1)$. The price level is P_t , gross inflation $\Pi_{t+1} \equiv P_{t+1}/P_t$, and $\pi_{t+1} \equiv \log \Pi_{t+1}$. The only nominal asset is a one-period nominal bond with price q_t^n , $R_t^n = 1/q_t^n$, $i_t = R_t^n - 1$; the net real rate is r with $1 + r = 1/\beta$.

1.1 (10 pts) The central bank pegs the nominal rate, $R_t^n = R^n$. Show that expected inflation is pinned down but the price level is not. A colleague argues: “to hit a price-level target P^* , just commit to the peg consistent with P^* .” Explain precisely why no *constant* peg can deliver P^* .

1.2 (14 pts) Replace the peg with a rule that reacts to the (log) price level, $i_t = r + \phi(p_t - p^*)$ with $\phi > 0$ (a *Wicksellian* rule), where p^* is a target and $p_t \equiv \log P_t$. Working in logs ($i_t = r + \mathbb{E}_t p_{t+1} - p_t$), derive the law of motion for p_t , find the values of ϕ for which the price level is determinate, and state what it selects. Is the Taylor principle ($\phi > 1$) required?

1.3 (6 pts) Contrast what this Wicksellian rule pins down with what an inflation rule $i_t = r + \phi\pi_t$ ($\phi > 1$) pins down.

Problem 2. Barro tax smoothing with non-contingent debt (30 points)

A government with convex tax-distortion costs smooths taxes (Barro): it raises flat-rate revenue τ_t and trades one-period real debt at constant gross rate $1 + r = 1/\beta$, starting from zero debt ($b_0 = 0$). Normal spending is $\bar{g}$ every period, but a *war* adds $\Delta > 0$ to spending for the N periods $t = 0, 1, \dots, N - 1$, after which spending returns to $\bar{g}$ forever.

2.1 (12 pts) Show the optimal tax rate is constant and compute it for this spending path. Verify that a one-period war ($N = 1$) reproduces the familiar “temporary shock” result, and describe how τ varies with the war’s length N .

2.2 (10 pts) After the war ends, is the budget balanced? Compute the long-run (post-war) debt level and the primary surplus that services it.

2.3 (8 pts) Interpret: which part of the war is debt-financed versus tax-financed, and what changes if the war’s duration is uncertain?